

AI-Based Anomaly Detection for Predictive HACCP Using IoT-Based Environmental Monitoring

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Module Title

Individual Project

Abstract

Manual food safety monitoring within Hazard Analysis Critical Control Points(HACCP) frameworks remains largely reactive. Surface swabs are taken at intervals, laboratory results take up to 72 hours to return, and by that point product has often already left the facility. This delay leaves food manufacturers with limited ability to intervene before a contamination risk becomes a real problem. This project proposes an AI-based anomaly detection system. It combines continuous IoT environmental monitoring with historical industrial swab records to support earlier and more reliable identification of contamination risk. Two DHT22 sensors and a reed switch door sensor will feed real-time temperature, humidity, and access data into Microsoft Azure for secure cloud-based processing. Three machine learning models Random Forest, LSTM, and Isolation Forest will be developed and compared using precision, recall, F1-score, and detection delay, with SHAP analysis applied to interpret each model's predictions. The study contributes a rare example of AI anomaly detection validated against genuine industry swab data rather than simulated results. In practical terms, a system like this could give food manufacturers an early warning capability that current manual processes simply cannot provide.

Context, Rationale and Research Gap

Food safety is a serious public health issue. The World Health Organization (2024) estimates that about 600 million people fall ill due to contaminated food, with about 420,000 deaths recorded worldwide. In the UK, *Listeria monocytogenes* is one of the hardest pathogens to manage in food production. It can tolerate cold temperatures and thrives in damp conditions, making it hard to eliminate from the surfaces of food factories (Food Standards Agency, 2023). HACCP is the primary method of managing risks associated with food manufacturing. In practice, most sites still rely on staff taking swabs at fixed intervals and waiting one to three days for lab results. By then, the product has often already left the building, so problems get found after the event rather than stopped before it.

Recent studies explore how AI and IoT technologies might address these limitations. Work by Kanicheril Ambikalekshmi et al. (2025) and Mu et al. (2024) shows that machine learning models fed with live temperature and humidity readings can flag warning signs well before a manual check would catch them. This project will use Microsoft Azure to handle the data processing side.

What is still missing from the research is real factory swab data being used to test these systems. This project directly addresses that gap by combining IoT sensor data with real industry swab records to validate AI-based anomaly detection in an operational food manufacturing environment.

Table 1 Comparison of Traditional HACCP and Proposed AI-Based Monitoring

Feature	Traditional HACCP	This Project
Data Collection	Manual swabbing	IoT environmental sensors
Reaction Time	24–72 hours	Real-time monitoring
Approach	Reactive	Predictive
Data Source	Periodic sampling	Continuous sensor + swab history

Aim and Objectives

Aim

This project aims to build and test an AI-based system that uses IoT sensor data together with factory swab records to support HACCP monitoring in food production. The project will examine whether continuous automatic monitoring can detect possible contamination earlier than the current manual swab process.

Objectives

Literature Review and Data Preparation (Weeks 1–6)

To review recent research on AI anomaly detection, IoT monitoring, and HACCP. At least 20 academic sources will be studied. During this time, the required datasets will also be collected and cleaned, aiming for at least 90% data completeness. Industrial swab data from a UK cheese factory will be used if approved, with public datasets available as backup.

IoT System Setup (Weeks 6–10)

To build a Raspberry Pi system using two DHT22 sensors and one door sensor. The system will send temperature and humidity data to Microsoft Azure every minute, with minimal data loss over two weeks.

AI Model Development (Weeks 10–16)

To train and compare three machine learning models for anomaly detection. Their performance will be calculated based on some common parameters such as precision, recall, and F1-score.

Evaluation and Reporting (Weeks 16–20)

This part will be for comparing the results obtained from the AI with the results obtained from the manual inspection, and whether the AI can be practically implemented in the factory. The findings will be clearly presented and discussed in the final dissertation.

Table 2 SMART Summary of Project Objectives

Objective	Specific	Measurable	Achievable	Relevant	Time-Bound
1. Literature & Data	Review 20+ sources and prepare datasets	90% completeness target	Standard MSc research scope	Informs model design and research gap	Week 6
2. IoT Development	Deploy Raspberry Pi with 2×DHT22 + reed switch	≥1 reading/min, <5% data loss	Low-cost hardware + Azure student account	Provides live environmental data	Week 10
3. AI Modelling	Train RF, LSTM, Isolation Forest	≥85% accuracy, precision/recall/F1, SHAP analysis	Standard ML libraries available	Core technical contribution	Week 16
4. Evaluation	Statistical comparison + feasibility framework	p < 0.05 significance	Realistic within the final project phase	Answers research questions	Week 20

Justification

This project is suitable for the MSc level because it is not simply about running a machine learning algorithm. It brings together knowledge from across the programme, machine learning, data mining, artificial intelligence, and IoT systems to build one complete system tested against real industrial swab records rather than simulated data. That practical grounding is what makes it more demanding than a typical undergraduate project.

Several modules feed directly into the work. Machine learning and data mining underpin the training and comparison of three models. AI concepts guide the anomaly detection approach and the interpretation of results through SHAP analysis. IoT knowledge drives the Raspberry Pi sensor setup, and cloud computing skills are applied through Microsoft Azure for real-time data handling. No single module covers this ground alone; the project genuinely requires combining everything learned across the programme.

The benefits are both practical and academic. Food manufacturers could use a system like this as an early warning tool, helping to reduce contamination incidents and costly product recalls. Food safety professionals may find value in seeing how AI can work alongside existing HACCP practice rather than replace it. For the research community, testing anomaly detection models against genuine factory swab records is still relatively uncommon, and this project adds directly to that evidence base.

The work is challenging, hardware deployment, cloud integration, data preparation, and model evaluation all need to happen within a tight timeline. It remains achievable because the hardware is affordable, the tools are well established, fallback datasets are in place, and four years of hands-on experience in food manufacturing hygiene means the design decisions throughout are grounded in real operational knowledge.

Methodology, Approach and Resources

This project follows a quantitative experimental research design. A Raspberry Pi prototype fitted with two DHT22 temperature and humidity sensors at separate critical control points and one reed switch door sensor will collect continuous environmental readings every five minutes. Data will be transmitted to Microsoft Azure IoT Hub for secure cloud-based ingestion and storage. Azure ML will be used for model training and deployment where appropriate. All data will be cleaned, timestamp-aligned, and checked for completeness before analysis. Swab results will be time-aligned with the 24-hour sensor window preceding each sample collection to identify pre-contamination environmental signatures. This approach directly links continuous IoT readings to real microbiological outcomes.

The primary dataset will be industrial swab records from a UK cheese manufacturing facility, subject to ethics approval. If access is delayed or denied, FSA Food Hygiene Rating Scheme data and UCI environmental sensor datasets are confirmed fallback sources.

Three models will be developed and compared. Random Forest will handle interpretable classification of combined sensor and swab data. LSTM will detect temporal patterns across continuous sensor streams. Isolation Forest will identify anomalies without requiring labelled failure examples. Each model will be evaluated using precision, recall, F1-score, and detection delay, defined as the time between the first environmental anomaly and the corresponding positive swab result. An ablation study will test the contribution of individual sensor inputs, and SHAP analysis will explain which variables drive each prediction.

Table 3 Project Resources

Category	Details
Hardware	Raspberry Pi 4, 2x DHT22, reed switch
Software	Python, scikit-learn, TensorFlow, Jupyter, GitHub
Cloud	Azure IoT Hub and Azure ML free student tier
Data	Industry swab records pending, FSA confirmed, UCI confirmed

Ethics & Data Management

This project uses industrial swab records and environmental sensor data. Before anything is analysed, all company data will be anonymised and stripped of anything that could identify the business or its staff. No personal employee information will be collected at any stage.

The research will only use data that is genuinely needed for the study. If the company agrees to share their records, a formal data agreement will be put in place first, and the data will be used for this project only. Everything will be stored on password-protected devices and through university-approved systems. Day-to-day access stays with the researcher, and the supervisor will be brought in if any decisions around the data need a second opinion.

University ethical approval will be requested before any company data is touched. The FSA open data and UCI datasets used as fallback sources are publicly available and need no additional approval. The supervisor will be kept informed throughout to make sure the project stays within ethical and legal boundaries.

Risks & Mitigation Plan

Every project carries some degree of uncertainty, and this one is no different. The risks below have been identified based on the practical realities of working with industry data, low-cost hardware, and a fixed submission deadline. Each one has a realistic mitigation plan in place so that progress can continue regardless of what happens.

	Impact					
Likeli hood		Negligible	Minor	Moderate	Major	Catastrophic
	Very low	Low	Low	Low	Medium	High
	Low	Low	Low	Medium	Medium	High
	Medium	Low	Medium	Medium	High	High
	High	Medium	Medium	High	High	Extreme
	Very High	Medium	Medium	High	Extreme	Extreme

Figure 1 Project Risk Assessment Matrix

Table 4 Identified Risks and Mitigation Strategies

Risk	Likelihood	Impact	Risk Level	Mitigation
The company denies or delays data access	Medium	Major	High	FSA open data and UCI datasets confirmed and ready as an immediate fallback
IoT sensor failure or data loss	Low	Moderate	Medium	Spare components ordered in advance, synthetic temperature and humidity data generated from historical patterns used as a substitute
Azure integration is more complex than expected	Medium	Moderate	Medium	Core models built and tested locally first, Azure added as an extension
ML models do not reach the targeted performance threshold	Low	Moderate	Medium	Three models compared findings remain valid as a comparative study, regardless of individual accuracy levels
Time pressure from competing deadlines	Medium	Major	High	Weekly milestones set from the start, dissertation writing runs alongside development

Ethics approval delayed	Low	Major	Medium	Open datasets are used immediately while approval is pending, and no company data is touched until cleared
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Gantt Chart

Phase	Activity	Deliverable	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Planning	Proposal Submission	Due 3 Mar								
Literature	Literature Review									
Ethics & ToR	Ethics Form	Due 15 May								
Ethics & ToR	Terms of Reference	Due 22 May								
IoT Dev	Hardware Setup & Azure Config									
Data	Data Collection & Preprocessing									
ML Models	Random Forest									
ML Models	LSTM									
ML Models	Isolation Forest									
Evaluation	Model Evaluation & SHAP									
Writing	Dissertation Writing									
Submission	Final Submission & Presentation	Due 1 Sep								

Figure 2 Project Timeline (Feb–Sep 2026)

Staff Consulted

Dr Ann Smith, Project Supervisor

References

- Food Standards Agency. (2023). *Listeria monocytogenes: Guidance for food business operators*. <https://www.food.gov.uk>
- Kanicheril Ambikalekshmi, A., Eruaga, M., & Saha, S. (2025). Advancing food safety through artificial intelligence: A detailed review of monitoring and detection technologies. *Food Safety and Health*. <https://doi.org/10.1002/fsh3.70050>
- Mu, W., Bouzembrak, Y., Marvin, H., & van der Fels-Klerx, H. J. (2024). Making food systems more resilient to food safety risks by including artificial intelligence, big data, and internet of things into food safety early warning and emerging risk identification tools. *Comprehensive Reviews in Food Science and Food Safety*. <https://doi.org/10.1111/1541-4337.13296>
- Naseem, S., & Rizwan, M. (2025). The role of artificial intelligence in advancing food safety: A strategic path to zero contamination. *Food Control*. <https://doi.org/10.1016/j.foodcont.2025.001>
- Revelou, P. K., Tsakali, E., Batrinou, A., & Strati, I. F. (2025). Applications of machine learning in food safety and HACCP monitoring of animal-source foods. *Foods*, 14(6), 922. <https://doi.org/10.3390/foods14060922>
- World Health Organization. (2024). *Food safety: Key facts*. <https://www.who.int/news-room/fact-sheets/detail/food-safety>